H. ASHEN KAVISHA

Information Technology Undergraduate (specialized in Data Science)

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Ashen Kavisha

Ashen Kavisha Lakshan

Portfolio

Profile Summary

Third-year IT undergraduate at SLIIT specializing in Data Science, with hands-on experience in full-stack web development, mobile applications, and machine learning. Built and deployed production projects including a live freelance platform and an ML classification system for lung cancer prediction. Proficient in Python, React Native, Node.js, and MongoDB, with a strong foundation in data preprocessing, feature engineering, and model training. Passionate about building scalable, data-driven applications that solve realworld problems.

Main Projects (Click the titles to access the projects)

NASA APOD Gallery (ASP.NET Core MVC)

(Aug 2026 – Sep 2026)

Tech Stack: ASP.NET Core MVC, C#, ADO.NET (SqlClient), SQL Server, Bootstrap

- Built an ASP.NET Core MVC web app that fetches astronomy pictures from NASA's APOD API using HttpClient and saves them to SQL Server.
- Used ADO.NET (SqlClient) to store and read entries, skipping duplicate dates so repeated fetches don't create duplicate records.
- Added validation to filter out invalid API data, and a fallback to NASA's Image and Video Library when APOD returned unusable results.
- Handled broken image links with a placeholder image so the gallery never shows empty cards.
- Read saved records from the database and displayed them in a Bootstrap image gallery.
- Kept the real API key out of the public GitHub repo using local config files.

Retail Banking Data Warehouse & Business Intelligence Solution

(Aug 2026 – September 2026)

Tech Stack: SQL Server , T-Sql ,Python,Pandas,Power BI ,ETL

- Implemented ETL workflows to extract,clean, transform and load banking data into a dimensional data warehouse.
- Developed SQL data marts for monthly transactionanalysis and district-level loadn performance reporting.
- Implemented incremental fact loading and surrogate-key mapping to maintain data integrity and prevent duplicate records.
- Performed data validation and reconciliation, ensuring consistency across **1,056,320 transactions and 682 loan records**.
- Collaborated with team members to integrate the dimensional warehouse and prepare validated analytical datasets for Power BI dashboards.

Lung Cancer Prediction System (**Machine Learning)**

(July 2025 – November 2025)

Tech Stack: Python, Pandas, NumPy, Scikit Learn, Seaborn

- Developed ML classification system with optimized feature engineering pipeline using SelectKBest and mutual information scoring, achieving dimensionality reduction from full feature set to top 10 predictive features.
- Implemented end-to-end data preprocessing workflow including class balancing,normalization and correlation analysis (threshold>0.85) to ensure high-quality training data for medical prediction tasks
- Built reproducible Python pipeline using Scikit-learn, Pandas, and Seaborn for automated feature selection and statistical analysis of patient health data

Assignment service web application

(December 2025 – February 2026)

Tech Stack: Java, Spring Boot, React, PostgreSQL, AWS EC2, Nginx, Vercel

- Created an assignment service platform for submit assignments based on user needs.
- Users can submit assignments , and get solution delivery via email and users dashboard notification inbox.
- Created an admin panel to receives the assignments that submit by users. And it receives via email to the administrator .
- Administrator approve the assignment and send the solutions to the user via email, user heads to the payment gateway and pay for the service.
- Deployed the production application across Microsoft Azure, AWS EC2, and Oracle Cloud, gaining hands-on experience with multiple cloud environments and migrating between providers based on infrastructure cost and operational requirements.
- Configured production infrastructure: Nginx reverse proxy, Let's Encrypt SSL, cross-subdomain cookie authentication, and automated deployment workflow (GitHub → EC2 → systemd restart).
- Used PostgreSQL for data persistence with secured, environment-based credential management (no hardcoded secrets)
- Added security hardening including reCAPTCHA v3 and rate limiting on public-facing forms.
- Web site is now live take a look at - https://www.assignmentservice.net

E-Library Mobile App

(January 2026 – June 2026)

Tech Stack: React Native,Expo SDK,NODE + EXPRESS.js ,MongoDB,JSON

- This application is a cross platform mobile application that allows read books to the users.
- Created an favourite bookshelf to access the saved books.
- Users can manage a personal bookshelf with reading lists(Reading now, Favourites , Wishlist)
- Favourites individual books for quick access
- Also this contains AI powered book recommendations .
- USED Min-Max normalisation — puts rating and year on the same 0–1 scale as the similarity score so they can be added together fairly
- Also added Weighted score fusion — the core formula that combines all three signals (similarity, rating, recency) with tunable weights. And Metadata filtering — hard rules that remove books before ranking even begins (rating, year, author, language)

IPO Listing Profitable Prediction

(FreeTime(Own) Project Jun-2026)

Tech Stack: Python – TensorFlow / Keras , Pandas , Numpy, Sckit-Learn

- This Project Contains A Indian IPO would list profitably, using a dataset of ~320 IPOs. (Small Dataset).
- Removed Outliers Using IQR Clipping and Min-Max Scaling to balance the data that most important to change behaviour of Target Feature.
- Also added correlation heatmap so can identify which feature affects target feature. And less affect features removed.
- Splitted the dataset and achieved 72% test accuracy; also got the idea that accuracy can be increase via adding more data.
- Gained hands-on experience in the deep learning workflow: architecture design, regularization trade-offs, and the relationship between data quality/quantity and model generalization.

Bank Marketing: Term Deposit Prediction and Call Planner (Machine Learning)

(Sep-2026 – Oct 2026)

Tech Stack: Python, Pandas, NumPy, Scikit-learn, XGBoost, Streamlit, Git

- Worked in a team of four to predict which bank customers will subscribe to a term deposit, using the UCI Bank Marketing dataset (45,211 records, about 12% positive). I owned the feature engineering and preprocessing stage.
- Engineered 8 new features from 15 raw columns, such as previous-campaign success, loan count, and age, balance and contact-frequency groups.
- Built a Scikit-learn pipeline (imputation, scaling, one-hot encoding) and used SelectKBest with mutual information to cut 68 encoded columns to the 30 most informative, fitted on training data only to avoid data leakage.
- Saved the pipeline so my teammates trained their models on it, and the live app applies the same steps to new customers.
- The team's tuned XGBoost model reached an F1 of 0.465 on unseen data, and the top-ranked 10% of customers subscribed about 4x more often than average.
- Deployed the app on Streamlit Community Cloud, Live Demo: https://ashenkavisha-bank-marketing-call-planner.streamlit.app

NEXTGENEDU AI–University Admission Analysis

(Group Project(2026 Aug – 2026 Oct)

Tech Stack: R,R Shiny,Python,Jupyter NoteBook,Git

- Developed and compared six regression approaches for predicting admission chance using 400 applicant records, evaluating performance with MAE, RMSE and R^2.
- Built an R Shiny application supporting individual predictions, bulk CSV uploads, adjustable feature weights and combined applicant ranking.
- Applied PCA to six standardized applicant features, retaining research experience separately; achieved test RMSE of 0.0613 and R^2 of 0.8126 with PCA regression.
- Implemented percentage/seat-based interview shortlisting, applicant radar profiles, downloadable reports and visual explanations of analytical findings.
- Deployed the application to shinyapps.io and maintained project code and analysis notebooks on GitHub.
- **Live Demo:** https://nextgenedu.shinyapps.io/nextgenedu/

CI/CD DevOps Pipeline

(June 2026 – July 2026)

Tech Stack: React/Vite, Node.js / Express, PostgreSQL, Docker, Jenkins, SonarQube, AWS EC2, Nginx

- Designed and deployed a fully automated CI/CD pipeline across three dedicated AWS EC2 instances, triggered automatically via GitHub webhooks on every push.
- Implemented a Jenkins declarative pipeline covering dependency installation, frontend build, static code analysis (SonarQube), Docker image builds, and automated remote deployment via SSH.
- Containerized a full-stack application (React frontend, Express backend, PostgreSQL database) using multi-stage Docker builds, reducing final frontend image size by serving static assets through Nginx rather than a full Node.js runtime.
- Configured Docker Hub as the image registry and built an automated build-push-pull-deploy workflow, eliminating manual deployment steps entirely.
- Used Nginx and Jenkins infrastructure constrains to execure scheduling and dis monitoring thresholds.
- Diagram: GitHub → Jenkins (build & test) → SonarQube (code quality gate) → Docker Hub → AWS EC2 (live deployment).

Education

2024 - Present
BSc (Hons) in IT specializing in Data Science Sri Lanka Institute of Information Technology(SLIIT)

2023 G.C.E A/L in Physical Science stream

- Physics-C
- Combined Mathematics-C
- Chemistry-C

- ***Certifications*** : The Full Stack Web Development Bootcamp 2023-MERN STACK ,Numpy For Data Science Real Time Experience

Core Skills

- ***Programming*** : Python , JAVA , JavaScript , C# ,Kotlin , C++,R
- ***CLOUD & DevOps*** : AWS , Azure , Docker , Jenkins , SonarQube , Nginx , Oracle Cloud
- ***Web*** : Node.js , React , React Native / Vite, Express.js , Next.js , HTML , CSS, Bootstrap , WordPress
- ***Database*** : MySQL , SQL , MongoDB , MsSql ,PostgresSQL
- ***Tools & Others*** : Github , Git, Postman , Github Actions , Github Webhooks , SpringBoot , TomCat , Adobe Photoshop, Capcut , Adobe Premiere Pro ,DBeaver, PowerBI, Tableau , Gsoc open source, R studio
- ***Languages*** : English ,Sinhala